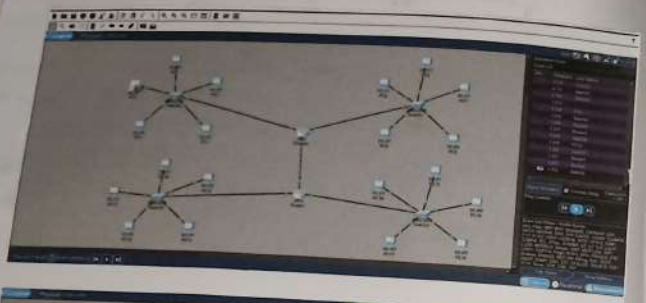
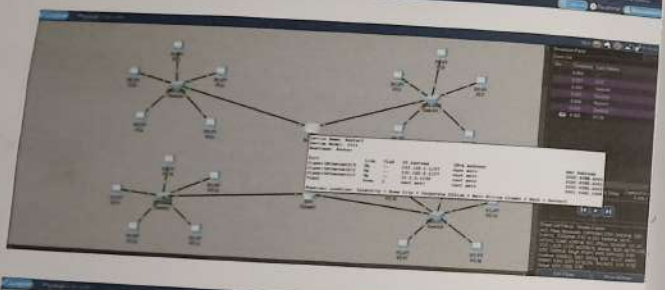
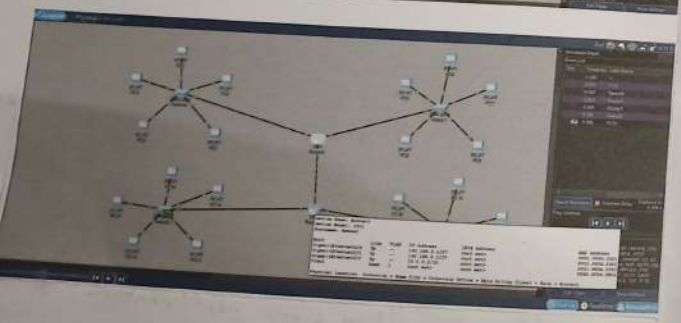






Ex.No.9 Implementation of Subnetting in Cisco Packet Tracer Simulator

Aim:

To implement subnetting in Cisco packet Tracer Simulator by creation multiple subnets, assigning static IP address, configure routs and establishing communication between different subnet.

Procedure:-

1) Open Cisco packet tracer and create a new network topology.

2) Add 2 router, 4 switches and 2PCs and connect them using appropriate cables

3) Consider the network 192.168.10.0/24 and divide it into smaller subnet using the /27 subnet mask 255.255.255.224

4) Configure Router0
   - G0/0 → 192.168.1.1/27
   - G0/1 → 192.168.20.1/27

5) Configure Router1:
   - G0/0 → 192.168.30.1/27
   - G0/1 → 192.168.4.1/27
   - G0/2 → 10.0.0.2/30

6) Enable all router interfaces using the no shutdown command

7) Assign static Ip address to the pcs

LAN1 : 192.168.1.11 - 192.168.1.15

LAN2 : 192.168.2.11 - 192.168.2.15

LAN3: 192.168.3.11 - 192.168.3.15

LAN4: 192.168.4.11 - 192.168.4.15

8) Assign the subnet mask 255.255.255.224 and the appropriate gateway to each pc.

9) Configure static router between router 0 and Router 1

Router 0:

ip route 192.168.3.0 255.255.255.224 10.0.0.2

ip route 192.168.4.0 255.255.255.224 10.0.0.2

Router 1:

ip route 192.168.1.0 255.255.255.224 10.0.0.1

ip route 192.168.2.0 255.255.255.224 10.0.0.1

10) Test communication between PCs in different subnets using the ping command.

11) Switch to simulation mode and sent a simple PDU between PCs from different subnets.

12) Observe the packet travelling through the switches and routers and verify successful communication

Result:

Thus, the subnetting was successfully implemented in Cisco packet tracer.